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Course code: DSA0604

Class Activity (20/07/2026)

Scenario 1: Hospital Patient Admission Analysis

```
library(ggplot2)

department <- c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency")
admissions <- c(120,95,150,110,180)

data <- data.frame(department, admissions)

print(data)

barplot(admissions,
        names.arg=department,
        col="skyblue",
        main="Patient Admissions by Department",
        xlab="Department",
        ylab="Admissions")

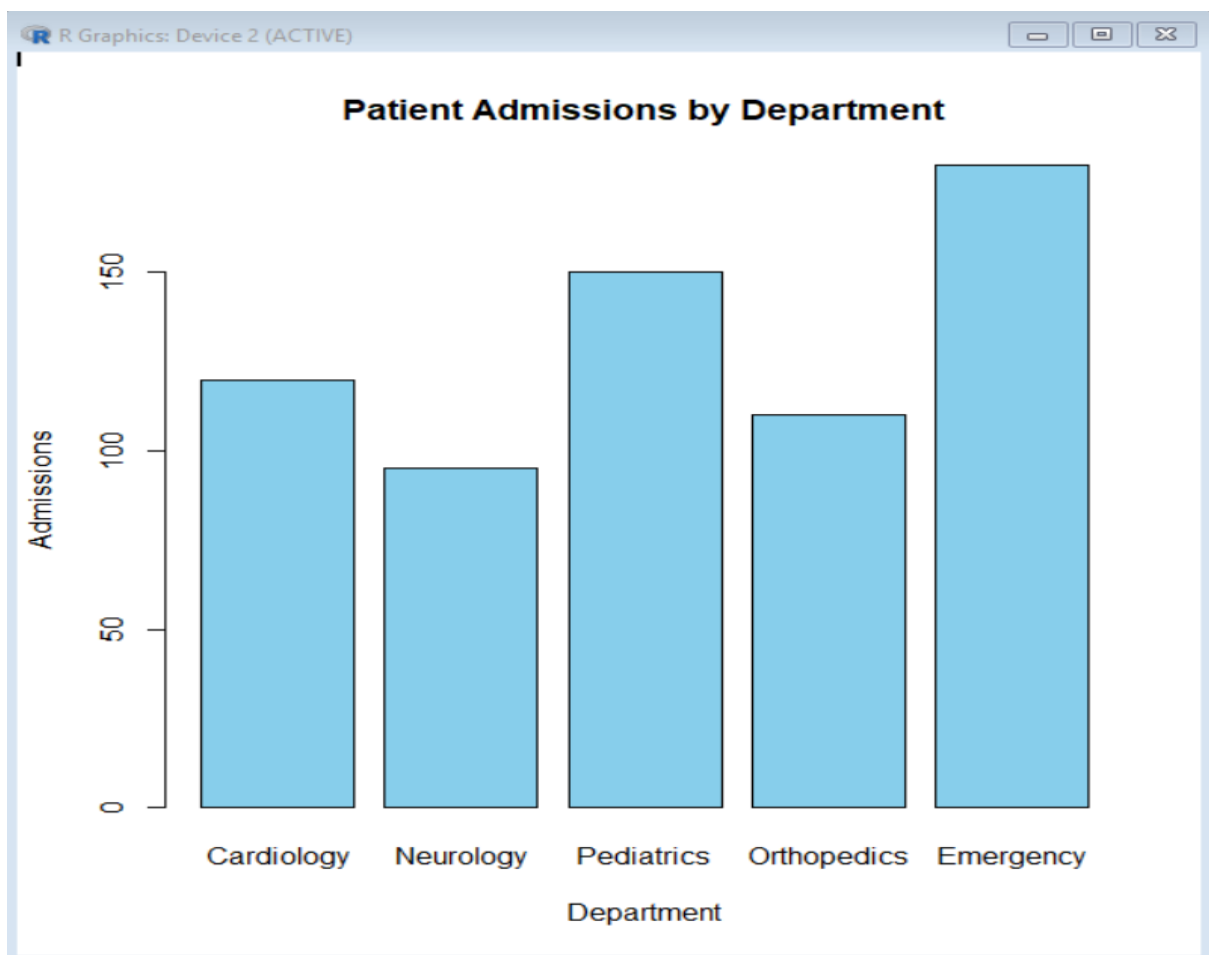
pie(admissions,
    labels=department,
    col=rainbow(5),
    main="Department-wise Admissions")

hist(admissions,
    col="lightgreen",
    main="Admissions Distribution",
    xlab="Admissions")

boxplot(admissions,
        col="orange",
        main="Admissions Box Plot",
        ylab="Admissions")
```

```
stack <- matrix(admissions, nrow=1)
barplot(stack,
        col=rainbow(5),
        main="Stacked Bar Chart",
        legend.text=department)
symbols(1:5,
        rep(1,5),
        squares=sqrt(admissions),
        inches=0.5,
        bg=rainbow(5),
        xlab="",
        ylab="",
        main="Treemap Representation")
```

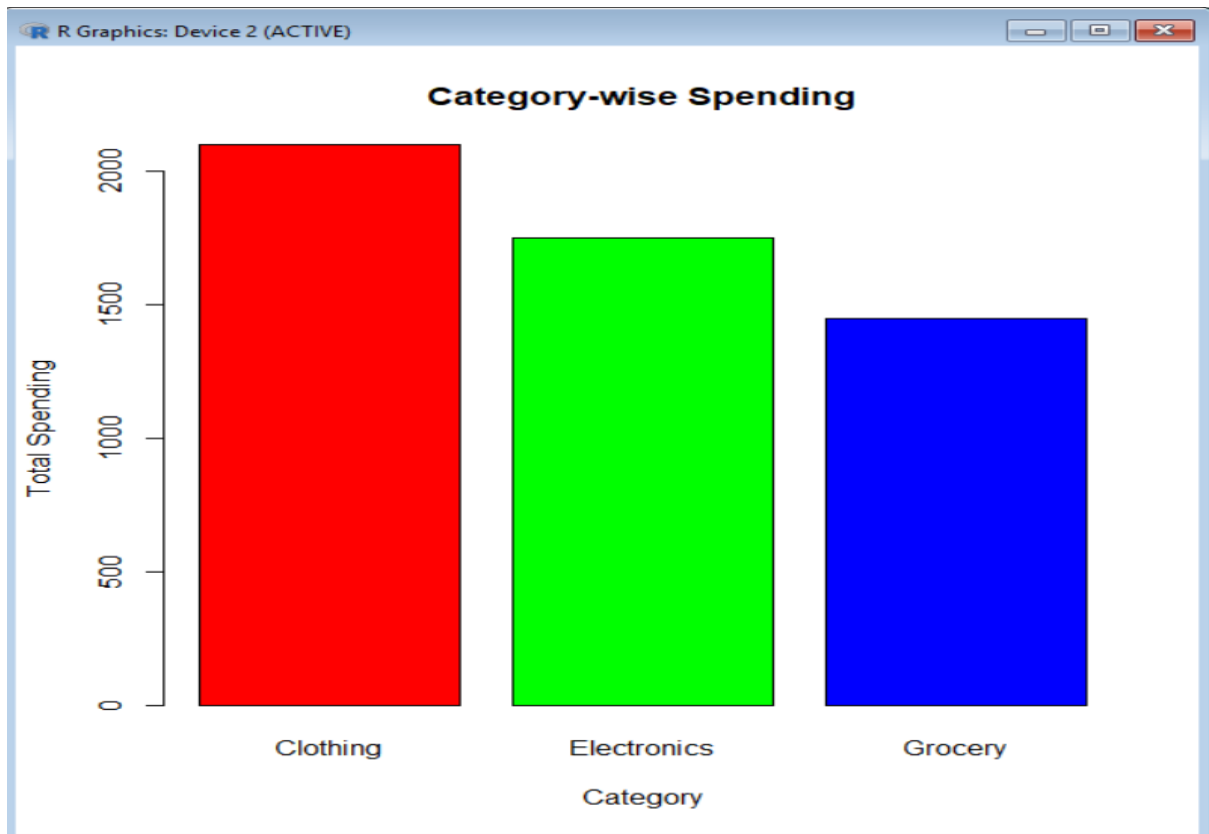
Output:



Scenario 2: E-Commerce Customer Spending Behaviour

```
category <- c("Electronics","Clothing","Electronics","Grocery",  
             "Clothing","Electronics","Grocery","Clothing")  
spending <- c(500,700,450,900,650,800,550,750)  
data <- data.frame(category, spending)  
print(data)  
hist(spending,  
     col="lightblue",  
     main="Customer Spending Distribution",  
     xlab="Spending")  
boxplot(spending,  
        col="orange",  
        main="Customer Spending",  
        ylab="Spending")  
plot(density(spending),  
     main="Density Plot",  
     col="blue",  
     lwd=2)  
total <- tapply(spending, category, sum)  
barplot(total,  
        col=c("red","green","blue"),  
        main="Category-wise Spending",  
        xlab="Category",  
        ylab="Total Spending")
```

Output:



Scenario 3: University Examination Performance Dashboard

```
subjects <- c("Maths","Physics","Chemistry","English","Computer")
```

```
marks <- c(85,78,92,88,95)
```

```
data <- data.frame(subjects, marks)
```

```
print(data)
```

```
barplot(marks,
```

```
  names.arg=subjects,
```

```
  col="skyblue",
```

```
  main="Subject-wise Marks",
```

```
  xlab="Subjects",
```

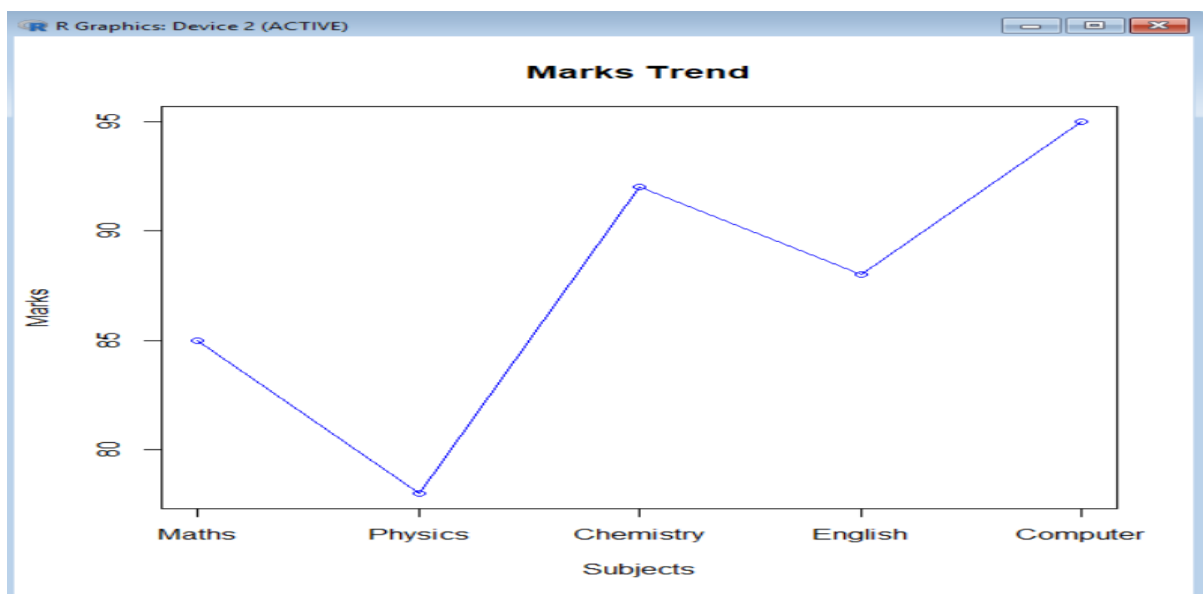
```
  ylab="Marks")
```

```
pie(marks,
```

```
  labels=subjects,
```

```
col=rainbow(5),  
main="Marks Distribution")  
hist(marks,  
col="lightgreen",  
main="Marks Distribution",  
xlab="Marks")  
boxplot(marks,  
col="orange",  
main="Marks Box Plot",  
ylab="Marks")  
plot(marks,  
type="o",  
col="blue",  
xaxt="n",  
main="Marks Trend",  
xlab="Subjects",  
ylab="Marks")  
axis(1, at=1:5, labels=subjects)
```

Output:



Scenario 4: Climate and Rainfall Distribution Study

```
months <- c("Jan","Feb","Mar","Apr","May","Jun")
```

```
rainfall <- c(45,60,80,120,160,200)
```

```
data <- data.frame(months, rainfall)
```

```
print(data)
```

```
barplot(rainfall,  
        names.arg=months,  
        col="skyblue",  
        main="Monthly Rainfall",  
        xlab="Months",  
        ylab="Rainfall (mm)")
```

```
plot(rainfall,  
     type="o",  
     col="blue",  
     xaxt="n",  
     main="Rainfall Trend",  
     xlab="Months",  
     ylab="Rainfall (mm)")
```

```
axis(1, at=1:6, labels=months)
```

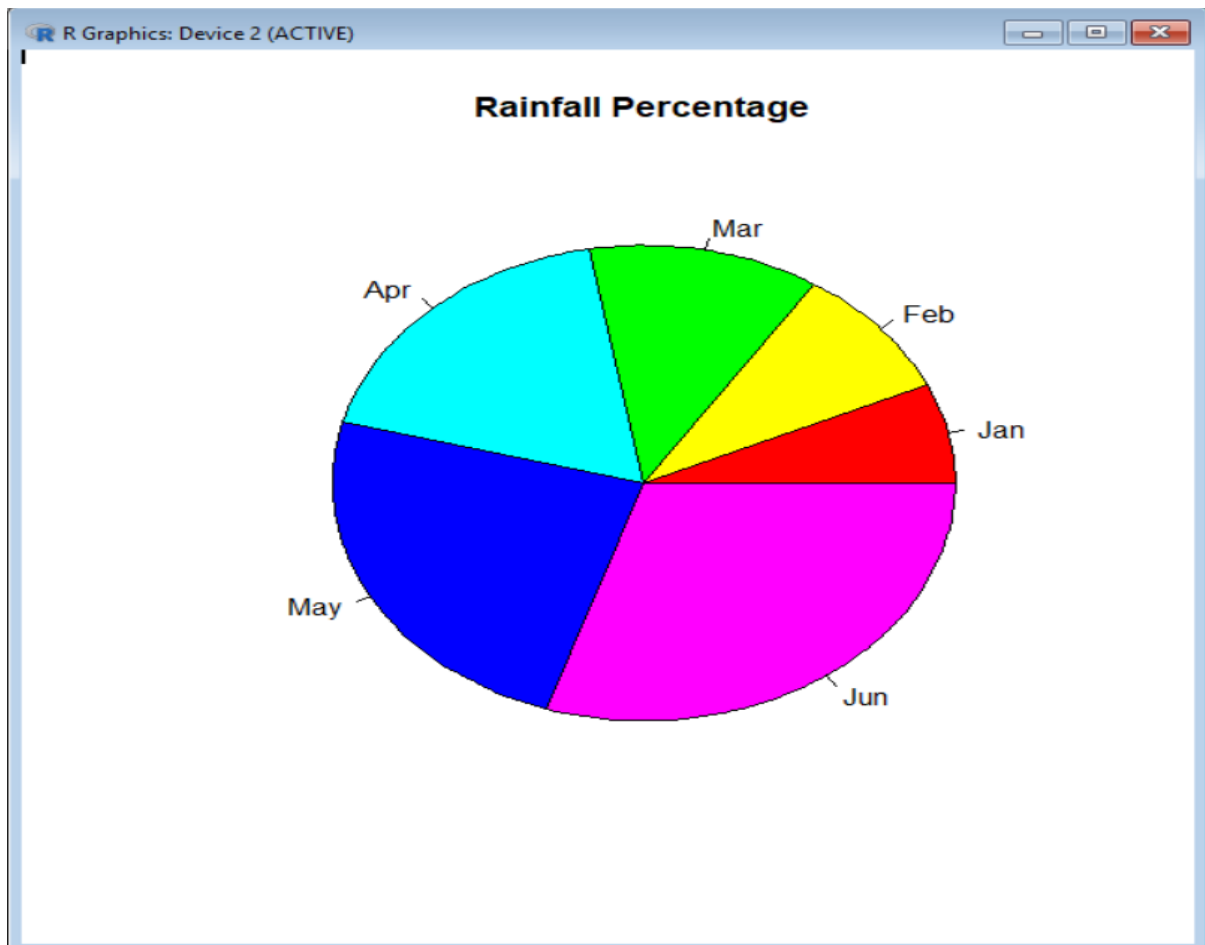
```
hist(rainfall,  
     col="lightgreen",  
     main="Rainfall Distribution",  
     xlab="Rainfall (mm)")
```

```
boxplot(rainfall,  
        col="orange",  
        main="Rainfall Box Plot",  
        ylab="Rainfall (mm)")
```

```
pie(rainfall,  
    labels=months,  
    col=rainbow(6),
```

```
main="Rainfall Percentage")
```

Output:



Scenario 5: Bank Loan Risk Assessment

```
customer <- c("C1","C2","C3","C4","C5","C6")
```

```
loan <- c(50000,75000,60000,90000,45000,80000)
```

```
risk <- c("Low","High","Medium","High","Low","Medium")
```

```
data <- data.frame(customer, loan, risk)
```

```
print(data)
```

```
risk_count <- table(risk)
```

```
barplot(risk_count,
```

```
col=c("green","yellow","red"),
```

```

    main="Loan Risk Categories",
    xlab="Risk Level",
    ylab="Number of Customers")
pie(risk_count,
    labels=names(risk_count),
    col=c("green","yellow","red"),
    main="Loan Risk Distribution")
hist(loan,
    col="lightblue",
    main="Loan Amount Distribution",
    xlab="Loan Amount")
boxplot(loan,
    col="orange",
    main="Loan Amount Box Plot",
    ylab="Loan Amount")
dotchart(loan,
    labels=customer,
    main="Customer Loan Amount",

```

Output:

